

Certify, Answer, Flag, Abstain: A Chain of Custody for Machine-Read Mathematics

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Abstract

A deployed reasoning system’s output should not be an answer; it should be a decision. We present a system that reads algebra-in-words into typed factor graphs, solves them exactly, and emits one of four decisions — **certify, answer, flag, or abstain** — through decision machinery that is entirely zero-parameter: no gradient flows through any component that produces a verdict. Behind a four-link chain of custody (input register, rendering invariance, lineage invariance, truth), the system certifies 912 of 1,500 held-out problems at measured 1.0000 precision — a zero-numerator bound, reported as such. The trained footprint is 8.0M parameters against a 506M-parameter frozen trunk slice; verification adds zero. On foreign benchmark prose the same channel initially certified garbage at 2% precision. We report this at full strength, because the zero-parameter recognition gate it forced now refuses 100% of that text: certification is distribution-bounded, and the bound is measured, gated, and published. A three-book hand-annotation campaign moved the reading frontier until its yield curve measurably saturated. Everything was built under registered predictions with mechanized verdicts across fourteen model generations; the complete ledger ships as supplementary material, and the paper closes with a falsifiable prediction about its own instrument’s aging.

1. Introduction

Deployed language models produce answers without knowing when they are right, and the discipline’s natural instinct — read the model’s own confidence — measures the wrong thing. We can state this as a measurement rather than a position. In our system, the entropy of a five-way vote across re-renderings of the same problem separates fragile parses from settled ones almost perfectly; it cannot separate settled-and-correct from settled-and-wrong ($H = 0.000$ versus 0.212 — the confidently wrong are nearly as quiet as the confidently right; Figure 2). **Temperature is orthogonal to truth.** Any architecture that ends at an internal confidence signal has built a depth gauge and called it a compass. This paper is about what to build instead.

The artifact is a **certification lattice** (Figure 1): a system whose output is one of four decisions — certify, answer, flag, abstain — produced by a chain of custody in which each link checks one invariance the previous link cannot. A recognition gate asks whether the input is in the register the system was calibrated on; a five-view vote asks whether the parse survives re-rendering; a cross-lineage panel asks whether it survives a change of training history; the answer key — present only in measurement, never in deployment — asks whether it is true. Every link is zero-parameter, and every link is justified in §6 by a named specimen that defeats the chain without it. The accountability arithmetic is §3’s census: 8.0M trained parameters leveraged against a 506M frozen trunk slice, with every component on the verification path adding exactly zero — there is nothing on that path for training pressure to corrupt. At freeze, the certified channel covers 60.8% of the widest

fixture at measured 1.0000 precision, with the zero-numerator arithmetic stated where the number is (§6.2, §10).

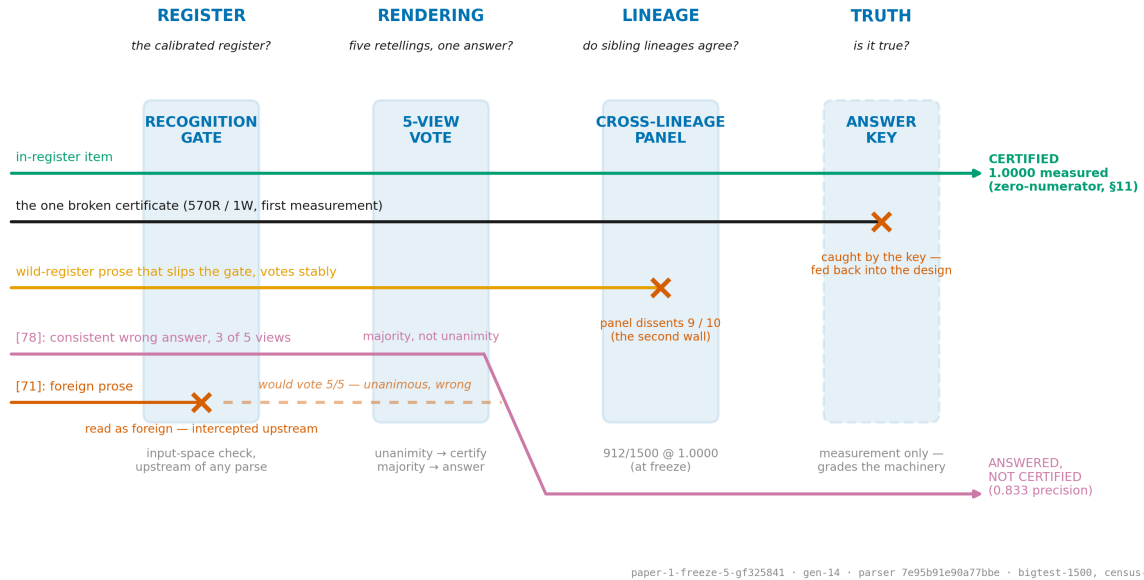


Figure 1. The chain of custody: four gates, four invariances (register, rendering, lineage, truth), five real trajectories — every gate annotated with the failure it provably catches.

The paper’s second product is the **method**. Every substantive claim began as a registered prediction with kill bars pinned before measurement; promotions are performed by battery scripts that either write the deployment manifest or touch nothing; and the complete chronological ledger — every registration, bar, verdict, and law — ships as supplementary material. It is long, unedited, and contains our mistakes at the same resolution as our results; that is what makes it evidence rather than narrative. §8 shows the discipline at full length on the campaign’s hardest problem, and §8.3 tables the laws it yielded, several of which (instrument rotation, estimator variance masquerading as distance, selection against promoted gates) we believe transfer to any measurement system that watches itself.

The **construction** is a reading campaign (§9): three books of real mathematical prose, hand-annotated through a gate the annotators could not flatter — it rejected its own author’s first five attempts — and measured to completion: the yield curve saturated (Figure 9), the mechanism was confirmed by three independent instruments, and the annotation rulebook was written entirely by the parser’s refusals.

The **limit** is reported at the same strength as the results. On a public benchmark slice with no author fingerprint anywhere on its input side, the certification channel initially signed foreign garbage — 2% precision where it measured 1.0000 in-distribution — and the flat abstention curve showed the system did not know what it did not know (§7). The recognition gate exists because of that measurement, now refuses 100% of the anchor’s false certificates, and the anchor sits permanently outside every acceptance path as the standing examiner. We claim no benchmark competence; we claim a certified channel whose boundary is measured, and §10 — drafted before any other section — is the recommended first stop for a skeptical reader: it states what the numbers do not show, including the zero-numerator bound, the $n=1$ -per-generation fact, and that the annotator is the system’s author.

The reader’s map: §2 situates the work; §3 derives the two-jaws architecture from the binding theorem; §4 gives the corpus discipline (including train/test disjointness verified up

to graph isomorphism); §5 measures the repair stack to its boundary; §6 presents the lattice; §7 the external anchor; §8 the method; §9 the reading campaign; §10 the honest limitations; contributions and author accounting close the paper.

2. Related Work

Every citation below was verified against its actual source before it was characterized, and the bibliography carries a per-entry note on what each work is cited for — the same use-matches-source standard the rest of the paper applies to its own numbers.

Selective prediction and abstention. The reject option is as old as pattern recognition (Chow, 1957; 1970), and selective classification’s risk-coverage framing (Geifman & El-Yaniv, 2017, and successors) is the natural coordinate system for our Figure 4. Two things distinguish the lattice from this line. First, the decision machinery itself is zero-parameter: nothing is trained on the accept/reject decision, so the selective mechanism cannot be shaped by the errors it will later be asked to catch — a property we argue for by named counterexample rather than by taste (§6.3). Second, the abstention literature typically evaluates a single trained model with a confidence head on one distribution; our claim is distribution-bounded by construction, with the boundary itself measured, gated, and reported as a headline result (§7) rather than as a threat to validity.

Calibration and internal confidence. Post-hoc calibration (Guo et al., 2017) and language models’ verbalized or self-evaluated confidence (Kadavath et al., 2022) all improve the readability of an internal signal. Our measurement problem with this entire family is §6.1’s: vote entropy — the strongest internal signal we found — cleanly separates fragile from settled parses and cannot separate settled-correct from settled-wrong ($H = 0.000$ vs 0.212). Internal signals read basin depth. We do not argue calibration is useless; we argue it calibrates the wrong axis for certification, which is why the lattice’s last two links are external (a differently-trained sibling and an answer key) rather than better-calibrated internals.

Self-consistency, met head-on. The closest relative of our vote is self-consistency decoding (Wang et al., 2023): sample multiple reasoning paths, take the majority. The mechanism differs in three load-bearing ways. Our five views are deterministic, solution-preserving re-renderings of the input (sentence permutations), not temperature samples — test-time augmentation in lineage (Krizhevsky et al., 2012; Shanmugam et al., 2021), repurposed from accuracy averaging to certification; notably, the dedicated study of TTA aggregation found naive averaging imperfect, and unanimity is not averaging. Agreement here measures invariance of the reading, not stability of a sampler. Unanimity is used as a certification tier with measured precision (1.0000 on 912, zero-numerator bound stated), not as an accuracy booster. And we report where the signal provably breaks: on out-of-register input all views are read by the same miscalibrated arm, agreement decouples from truth entirely (2% certified precision, §7), and a majority-vote system without an input-register gate inherits that failure silently. Self-consistency work generally reports the in-distribution win; §7 is the out-of-distribution invoice.

Conformal prediction. Conformal methods (Vovk et al., 2005; Shafer & Vovk, 2008; Angelopoulos & Bates, 2021) offer distribution-free coverage guarantees, which is more than our zero-numerator bound provides — under exchangeability between calibration and test data. That precondition is precisely what our external anchor violates and what the recognition gate exists to test: the mouth is, functionally, an explicit exchangeability check

run before any statistical claim is trusted. We view the two approaches as complementary — a conformal layer over the lattice’s scores is natural future work — but we note that our headline failure mode (§7.2, foreign text certified confidently) is exactly the one a conformal guarantee calibrated in-register would also have signed.

Propose/dispose architectures. The two-jaws design belongs to the family in which a learned proposer is checked by a sound verifier: semantic parsing of math word problems to equation forms (Zhang et al., 2020), autoformalization into proof assistants whose downstream proofs are machine-checkable (Wu et al., 2022), trained verifiers over sampled solutions (Cobbe et al., 2021; process-supervised in Lightman et al., 2023), and — in spirit — speculative decoding’s draft-then-verify asymmetry, where a cheap proposer’s output is accepted only by an exact check that preserves the target distribution (Leviathan et al., 2023; Chen et al., 2023). The lattice differs where §3.2’s census row does: our verification path contains zero trained parameters end to end (trained verifiers move the corruptible component rather than removing it), and abstraction is confined to the propose side by rule — macro-relations expand to primitives before the solver sees anything, so the checker’s soundness is never delegated.

Retransmission. §5’s repair stack is, deliberately, an ARQ system (Lin, Costello & Miller, 1984): error detection (the portfolio), negative acknowledgment (the flag), selective retransmission of flagged fields (the specialist), and a measured per-round recovery decay that motivates shallow retry budgets. We import the vocabulary because the communications literature solved the bookkeeping of when to stop re-asking long ago; our contribution there is the boundary measurement (§5, §8.2), not the loop.

Instrument aging. Distribution shift, deployment monitoring, and Goodhart-style dynamics — “any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes” (Goodhart, 1975; taxonomy in Manheim & Garrabrant, 2018) — all describe measures and detectors degrading in deployment. What we have not found articulated in the abstention or monitoring literatures is the selection mechanism of §6.4 — any signal promoted to a gate becomes selected against, because the surviving error population is by construction the population that passes it — together with its design consequence (a rotation portfolio that always holds one examiner out of the acceptance path) and a falsifiable, pre-registered prediction of the paper’s own detector decaying. We state this novelty claim carefully: the ingredients (Goodhart’s law, adversarial drift) are old; the articulation as a deployment law with a succession plan and a standing bet is, to our knowledge, new.

3. The Architecture (two jaws, derived)

3.1 The binding theorem, and the design it forces

The architecture is best introduced by the negative result that shaped it. A concept, in this domain, is not recoverable from either of its two surfaces. From the language side: problems that read as obvious kin — the taxi fare and the filling faucets of Figure 3, both creatures of the rate frame — share no graph class; their kinship is real, measured in the frozen trunk’s own geometry ($z = -2.05$), and entirely absent from their wiring. From the structure side: problems with identical canonical graphs — one digest, one knot — wear surfaces so different that no lexical method recognizes them as related. We call this the **binding theorem**: the concept is the binding between a linguistic frame and a structural role, and it lives in neither side alone. Figure 3 shows the four specimens; the structure-

side sweep that found graph-twins across fixture boundaries is §4’s isomorph audit, and both directions’ full measurement records are in the ledger.

Same frame, different knots

surface kinship is not graph kinship (the C2 split)

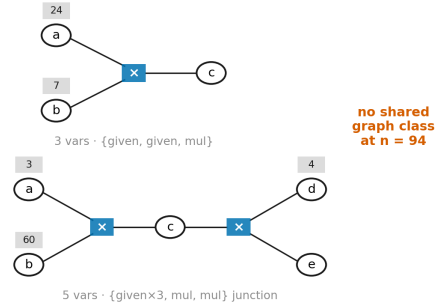
census [45] — “After traveling 50 miles by taxi, Ann is charged a fare of \$120. Assuming the taxi fare is directly proportional to distance traveled, how much would Ann be charged if she traveled 70 miles?”

RATE frame

same frame ($z = -2.05$ at the frozen trunk)

census [7] — “Three faucets fill a 100-gallon tub in 6 minutes. How long, in seconds, does it take six faucets to fill a 25-gallon tub? Assume that all faucets dispense water at the same rate.”

RATE frame



Same knot, different surfaces

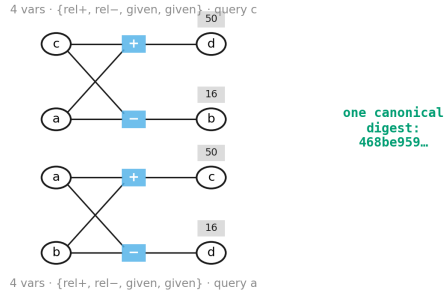
graph kinship is not surface kinship (the hash audit's cross-fixture isomorphs)

bigtest[1187] — “Let c, a, d, b be whole numbers. the fourth number has the value 16. The difference between c and a is b . Work carefully through each fact. d has the value 50. d is what you get from c plus a . What is c ?”

terse register

vtest[116] — “In this problem we are working with the numbers a, b, c, d . If a and b are gathered into one pile, that pile amounts to c . If you look at d , you will find that its value is exactly 16. It turns out that c works out to be 50 in the end. If you start from a and take away everything b has, what remains is d what value does a end up holding?”

verbose register



The concept is the binding between a linguistic frame and a structural role — recoverable from neither side alone.

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Figure 3. The binding theorem by specimen: same frame, different knots (top); same knot, different surfaces (bottom). All four items and both graphs are real, from the banked fixtures.

The two-jaws design is this theorem applied. If concepts are bindings, then recognition and abstraction must live where the binding is made — on the parse side, in the trained components that read language into structure. The factor graph they emit is deliberately **frame-free**: it records variables, relations, and quantities, and forgets that the problem was about taxis. The factor-graph representation itself is standard (Kschischang, Frey & Loeliger, 2001); what the theorem forces is where its content may come from. And verification inherits neither side: the symbolic solver receives only the graph, searches it exactly, and is graded by the dataset’s answer key. When higher-level abstractions enter (macro-relations proposed from recurring subgraphs), they expand to primitives before the solver sees anything — the key always grades in primitives. The slogan form: **neural proposes, symbolic disposes** — abstraction may live in annotation and recognition, never in verification.

This gives the pipeline a cleaner algebra. The predicate registry is a table of the solver’s atoms — each relation kind an irreducible pattern it understands natively, a prime. A parsed problem is then a factorization: the parser’s task is to express a composite stated in words as a product of registry primes, and the frame-free graph is that factorization written down. Macro-relations are composites admitted for compression, but they expand to primes before the solver sees them, because the answer key grades only in primes. The canonical digests of §4 are the same idea made rigorous — a canonical factorization of the graph — which is why disjointness up to isomorphism is disjointness of prime structure, not of surface text. Two problems share a knot when they share a factorization; strangers in words can be

identical in primes, and kin in words can be coprime. The registry’s counted frontier (§9) is, in this language, a list of primes not yet discovered.

3.2 The components at freeze, census-verified

The construction jaw is a small trained head over a large frozen eye. The **trunk** is the embedding table and first four layers of a pretrained 1B-parameter language model, used input-side only and never trained. The **parser head** reads the trunk’s states into a typed factor graph through two slot banks — 24 variable slots bound to letters positionally, 24 factor slots — with bilinear pointers from factor arguments to variables, a six-way factor typing plus an argument multiplicity bit, and most-significant-digit-first quantity decoding. (The six factor types are the parse-side surface, not the relation inventory: registry relations enter as solver-side predicates bridged onto these types — §3.3 — which is how double-digit relation kinds ride on a six-way head output.) The **repair specialist** is a second head of the same architecture, retrained each generation on the gate model’s organic failures and consulted only when the vote abstains. Around them sits the zero-parameter decision machinery of §6 (views, votes, panel join, recognition gate, flag), and beneath them the **solving jaw**: a general constraint-search core (arc consistency, forced-only commits, a predicate registry) containing zero learned parameters and zero domain-specific code in its core.

The parameter census (Table 1), re-run at the freeze tag against the deployed checkpoints rather than quoted from memory:

Census row	Parameters
Trained and deployed (parser + repair specialist)	8,005,722
Trained, whole system — both jaws (the solver adds zero)	8,005,722
Frozen and leveraged (trunk: embeddings + layers 0-3)	505,954,304

The second row equaling the first is the architecture’s claim in numbers: everything added to make answers checkable — search, verification, certification — added nothing trainable, so there is nothing on the verification path for training pressure to corrupt. The leverage ratio (63× frozen per trained; 126× against the parser alone) states the design bet: a pretrained model’s early layers already carry the reading; the trained head only learns where to point. Two sibling heads (4.0M and 13.8M, one differing in lineage and one in width) are additionally deployed at the certification tier as cross-examiners (§6.2); they produce votes, never answers.

3.3 As-built versus as-designed

The honest paragraph. The design that survived contact is narrower and better than the one first drawn. The predicate registry did what it was designed to do: relations enter as a predicate plus a parse-side bridge, with zero edits to the search core, and the registry has grown through double-digit relation kinds that way. Two designed components died: a hyperbolic embedding tier for the kind hierarchy was never needed — hard membership structures (masks, slots, letter positions) did its job with no learned geometry at all — and an elaborate memory/notebook mechanism gave way to a plain repair signal: the vote abstains, the specialist answers. In both cases a designed object was replaced by a measured action. The nouns died; the verbs survived.

3.4 Design laws as constraints, not lore

Three regularities from the campaign function as standing constraints on the architecture rather than commentary about it (sighting counts in Table 2). First, the **pointer law**: pointer errors are never fixed downstream of the pointer, so every new relation’s pointer is born candidate-restricted and span-supervised — the five remedies (masked attention, span supervision, a comma, alphabetical discipline, ballast) are a descending-cost toolkit applied at birth, not repairs applied after. Second, the **discovered dialect**: the annotation language the books converged on — consecutive letters, explicit knowns, one declarative relation per sentence — was never designed; it was written by the parser’s refusals, one rule per refusal, and it now functions as the system’s intermediate representation. The one designed logical form the project attempted is a tombstone (§9); the dialect that works was discovered under selection. Third, the **two-channel spine**: the strict separation of frame (parse-side) from structure (graph-side) was an early architectural guess that the binding theorem later proved load-bearing — collapse the channels and the concept has nowhere to live except entangled in both, which is precisely the failure the frozen trunk exhibits on its one chronic family (§10).

4. Corpus Discipline (how the training data cannot lie)

4.1 Solution-first, gate-refused

Every generated problem in this system is built backwards: a solution is constructed first, then a graph consistent with it, then text. The inversion is load-bearing because it makes correctness properties checkable at mint time rather than hoped for downstream. Two gates run on every candidate: a uniqueness gate (ban-and-resolve search under a fixed decision budget, where budget exhaustion rejects — an item whose uniqueness cannot be certified is not emitted) and a round-trip gate (the emitted text must parse back to a graph whose solution matches the one it was built from).

The design principle the gates enforce is that edge cases are made **unrepresentable rather than handled**. Three specimens, one principle: quadratic-family items are minted with perfect-square-by-construction discriminants, which dissolved the no-real-roots policy instead of implementing it; ill-defined selectors (a “largest” with no unique largest) self-gate as constraint violations and never reach text; and repeated-argument circuits carry an explicit no-give mechanism so the givens gate cannot leak the queried value. Three edge policies, zero new mechanisms — the generator’s grammar simply cannot say the broken thing.

4.2 Difficulty as measured axes, and the curriculum tombstone

Corpus difficulty is controlled by named, measured axes rather than vibes: teeth (rendering wildness — how far surface forms stray from canonical templates) and bands (structural depth — variables, chain length, relation mix). Both axes exist so that claims like “harder” have coordinates, and so that evaluation fixtures can be stratified by construction.

One honest sentence owed here: progressive difficulty ordering — the curriculum — is dead in this system at scale. It won its early ablation honestly (regime-tagged as such in the ledger), then inverted when the register mix widened: the schedule probe ran flat-mix against staged orderings from the same warm start and flat won outright. All training in the frozen stack is flat-mix; verdicts about orderings expire with their regime, and this one did.

4.3 The grading policy, measured against itself

Before any headline number was quoted, the grading policy was made uniform and then audited as an instrument. The uniform metric is forced-answer at the query variable (solution-set equivalence, not string match). Re-grading the then-current fixture under it: a raw 802 one-shot correct decomposed into 5 lucky-unforced answers (bounding the old metric’s luck inflation at **0.6%**) and 797 forced-correct. The audit’s real finding: **132 of 797 (16.6%) of correct answers come from graphs that differ from gold factor-wise** — right where asked, wrong somewhere unqueried. That ~17% equivalence class is stable across independent corpus draws (16.6/17.2) and is treated as a design parameter of the domain, not noise: it is why the certification chain (§6) grades answers through the solver rather than trusting graph match, and why “parse accuracy” and “answer accuracy” are never used interchangeably in this paper.

4.4 Disjointness up to isomorphism

The audit-swarm’s first question — does the test set leak into training? — is answered here at a stricter standard than string deduplication. Every problem is identified by a canonical Weisfeiler-Leman digest of its factor graph (Weisfeiler & Leman, 1968; Shervashidze et al., 2011): two problems with different letters, different surface text, and different generators that share a knot share a digest. WL-equivalence is coarser than exact isomorphism, so treating digest-equal items as identical is conservative for this purpose — the exclusion removes at least every true isomorph. Sweeping every fixture against every training corpus at this standard found **42 cross-boundary isomorphs** — items sharing a graph with something across a train/test boundary despite sharing no text (Figure 3’s bottom block shows one such pair). They were excluded, and the check is now a generation-bump gate: every promotion asserts train/test disjointness up to isomorphism before any battery number is read. We believe this standard should be ordinary; it is checkable in any system whose problems have canonical structure, and it is the difference between “we deduplicated” and “we know no knot is on both sides of the wall.”

5. The Repair Stack and Its Boundary

The thesis of this section: **the repair stack is measured to its boundary, and the boundary is a population, not a mystery.**

The anomaly portfolio. Two signals with opposite grammars watch every accepted parse: cross-view agreement (a dense ranker — it orders the whole population, best single AUC 0.840) and latent centroid distance (a rare flag — nearly silent, but precise where it fires). Their correlation is a modest 0.464, and combining them loses whole-ranking AUC while winning at every abstention operating point actually used (top-10% flags: kept-precision 0.862 vs 0.846) — the metric-must-match-decision-structure law’s fourth sighting, measured inside our own registration (Table 2).

Withhold-and-solve. When a parse is suspect, withholding the lowest-confidence factor and letting constraint propagation re-derive it recovers 26% of would-be errors for free — zero training, and zero silent-wrong introduced at any withhold depth, because a re-derived value that contradicts the graph refuses rather than guesses. This is the solver’s exactness used as a repair instrument: the deduction is only as available as the graph is redundant, and the mechanism prices exactly that.

Selective retransmission. The two-checkpoint design (§3.2’s parse head and repair specialist, one frozen trunk) lets flagged fields be re-asked rather than whole problems re-parsed. The clean result: field-level structural flags beat gold text-localization as repair conditioning — knowing which slot is wrong outperforms knowing which words are wrong, the structural-entry law’s cleanest demonstration, with the gold-leakage bound measured at zero. At its best measured operating point the specialist recovered 148 of 627 survivors of the one-shot pass.

The stack’s half-life. Repair rounds compose, and their yield decays hard: 19.6% → 7.7% → 1.1% → 0% across four rounds — a front-loaded shape observed four independent times (Table 2). The stack is therefore run shallow by design; composed end-to-end it lifted the then-current system to 47% on one domain and 32% on the other at convergence. Nothing in the stack pretends to iterate its way through a wall.

The boundary. What remains after the stack is a specific, counted population, and its character is the finding: the survivors’ internal states are 99.6% correctly decodable — the failure is a mis-aimed pointer, not a missing fact — and no downstream mechanism meaningfully re-aims a pointer (a perfect oracle ceiling of 13.9%; trained and input-side repairs in single digits; the arc’s full narrative and its nine registered kills are §8.2). The stack’s verdict on this population is detect-and-abstain, and its inheritance is the prevention constraint that now governs construction: every new relation’s pointer is born candidate-restricted and span-supervised, because the place to fix a pointer is before it is trained. The boundary did not end the repair story; it relocated it upstream.

6. The Certification Lattice

Temperature is orthogonal to truth.

6.1 The lattice as a decision structure

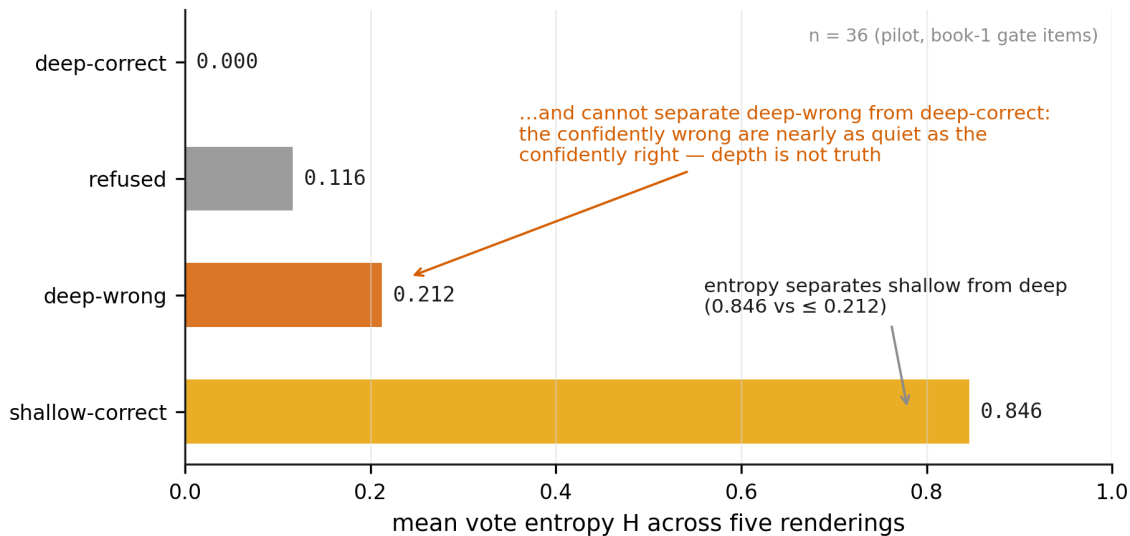
The system’s output is not an answer; it is one of four decisions — **certify**, **answer**, **flag**, or **abstain** — and the decision machinery itself is zero-parameter: vote counting, unanimity checks, a rank-sum read, an input-space distance threshold. The trained components — the parser and its repair specialist — produce candidate answers; they never produce verdicts. No parameter anywhere is trained on the certify/answer/flag decision, and nothing in the decision path consults the answer key: the key appears only afterward, as the grader of the machinery, never as a component of it. It is this decision-path purity, not an absence of learned parts, that the certification claims rest on.

The four rungs, with their dials as first measured:

1. **Certify** — five solution-preserving re-renderings of the input (sentence permutations), each parsed and solved independently; unanimity of the five forced answers. First measured at 0.9982 precision and 38.1% coverage.
2. **Answer** — majority vote (at least 3 of 5), with a specialist repair path on vote-abstain. Composite: 71.5% end-to-end at 0.833 precision.
3. **Flag** — a rank-sum of view-disagreement and centroid distance, read at the tail: kept-precision 0.862 at 10% abstention.
4. **Abstain** — no forced answer anywhere.

Behind the rungs stands a chain of custody, and each link answers a different question. The **recognition gate** asks is this input the kind of text the system was calibrated on? — an input-space check, before any parse. The **vote** asks is the parse invariant to how the problem is rendered? — five retellings must produce the same answer. The **cross-lineage panel** asks is the answer invariant to which training lineage produced the model? — independently trained siblings must agree. The **answer key** asks is it true? Register, rendering, lineage, truth: four invariances, in that order, and the sections below show by specimen why no link is redundant.

Why can't a confidence signal replace the chain? Because the most natural one reads the wrong axis. Vote entropy across the five renderings separates shallow parses from deep ones almost perfectly — in the pilot measurement ($n=36$), correct-but-fragile items read $H=0.846$ while deeply-settled items read near zero. But deeply-settled and correct are different properties: deep-correct items read $H=0.000$ and deep-wrong items read $H=0.212$ — the confidently wrong are nearly as quiet as the confidently right. Entropy measures basin depth, not truth. That is the epigraph in numbers, and it is the structural reason the lattice ends at an external key rather than at any internal temperature: no amount of introspective calibration substitutes for an invariance the system cannot fake.



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Figure 2. Vote entropy by outcome class ($n = 36$ pilot): entropy separates shallow from deep and cannot separate deep-wrong from deep-correct — temperature is orthogonal to truth.

6.2 The dials at freeze

At the frozen generation, the widest fixture (1,500 held-out problems) reads: 1195 answered correctly one-shot (79.7%); five-view unanimity certifying at measured 1.0000 precision; and the full cross-lineage panel — unanimity across five renderings and three models of distinct training histories (one differing in lineage, one in architecture width, with per-item behavioral disagreement measured rather than independence assumed) — certifying **912 of 1,500 (60.8%) at measured 1.0000 precision**. Figure 4 (the precision-coverage frontier) plots every rung: relaxing unanimity to 4-of-5 and 3-of-5 buys coverage at measured precision cost (0.9925 and 0.9832 at first measurement), and the frontier's history across generations shows the certification channel widening — from 0.9982 at

38.1% coverage in its first measurement to 1.0000 at 60.8% at freeze — as the reading campaign (§9) taught the parser the register rather than teaching it the test.

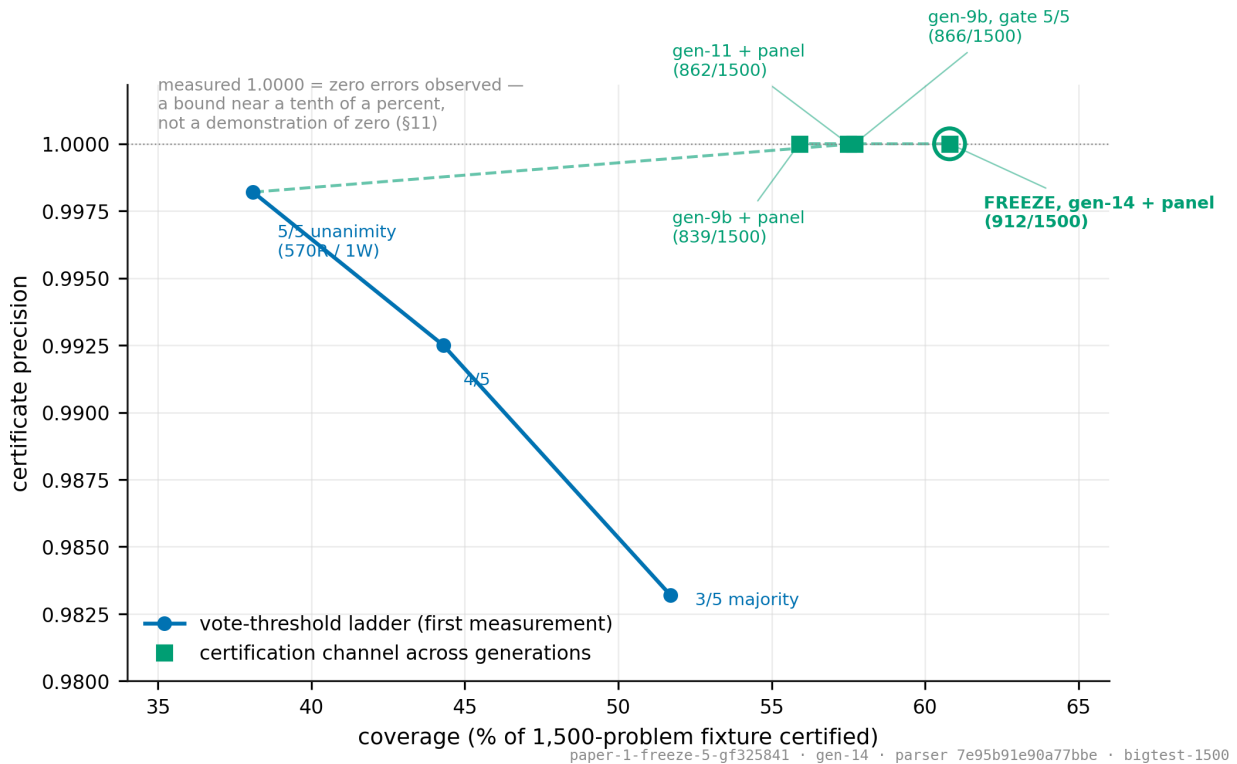


Figure 4. The precision-coverage frontier: the vote-threshold ladder at first measurement, and the certification channel widening across generations to the freeze point (912/1500 at measured 1.0000).

The statistical fine print is inherited from §10 and not repeated here: 1.0000 on 912 is a zero-numerator bound, not a demonstration of zero, and 1.5% of fixture items yield fewer than five distinct renderings. One number deserves emphasis precisely because it is not 1.0000: the channel’s first measurement contained a broken certificate (570 right, 1 wrong). The lattice’s history includes its own counterexample, found by the key, autopsied, and fed back into the design — which is the intended relationship between the machinery and its failures.

6.3 The specimens are load-bearing

Each link in the chain of custody has a named specimen showing what happens without it.

The false certificate exists. Problem [71] of the census fixture, presented as raw prose, voted 5/5 unanimous on a wrong answer — five renderings, one voice, wrong. This is the certification channel’s nightmare case, observed exactly where theory predicts it: on text outside the calibrated register, where all five renderings are read by the same miscalibrated arm and the vote’s independence assumption fails silently. The recognition gate exists because of this specimen — it reads the input’s register upstream of any parse, and it reads [71]’s prose as foreign. The doorman is not decoration; it is the link that intercepts the case the vote provably cannot.

The second wall. For prose that slips past the gate, the cross-lineage panel provides an independent check: on the census fixture’s raw-prose items where the gate model produced stable unanimous votes, the panel dissented on 9 of 10 (a later generation: 16 of 19). In-register, the panel is nearly idle — the single stable-wrong item in the 1,500-problem fixture

at the generation the panel was adopted was broken by cross-examination, 1 for 1. Its load-bearing jurisdiction is exactly the wild register: even text that slips the gate meets a jury drawn from a different training history. Renderings share a model’s blind spots; lineages do not.

The quiet failure shape. Problem [78]’s dialect voted a consistent wrong answer across three of five views — an answer-channel error that certifies nothing but answers wrongly with a stable voice. It is the specimen for why the answer rung and the certify rung carry different dials, and why deployment modes that cannot tolerate 0.833 must read from the certified channel only.

6.4 Instruments age, by mechanism

The lattice’s anomaly signals are instruments, and the campaign measured how they age rather than assuming they don’t.

The geometric monitor — per-kind centroids in the head’s latent space — lost discriminative power across generations. The audit found the mechanism, and it was not the monitor’s fault: the latent constellation rotates between generations. Raw cross-generation centroid cosines read ~ 0.59 , as if the spaces were unrelated; after orthogonal Procrustes alignment they read 0.988 with small residual. The constellation’s shape survives; its coordinates do not. The monitor aged because nobody told it the sky had turned. The repair is structural, not statistical: geometric instruments are re-anchored every generation as a standing duty, and cross-generation geometric comparisons are made only after alignment. (The panel is immune by construction — its votes are answers, not coordinates.)

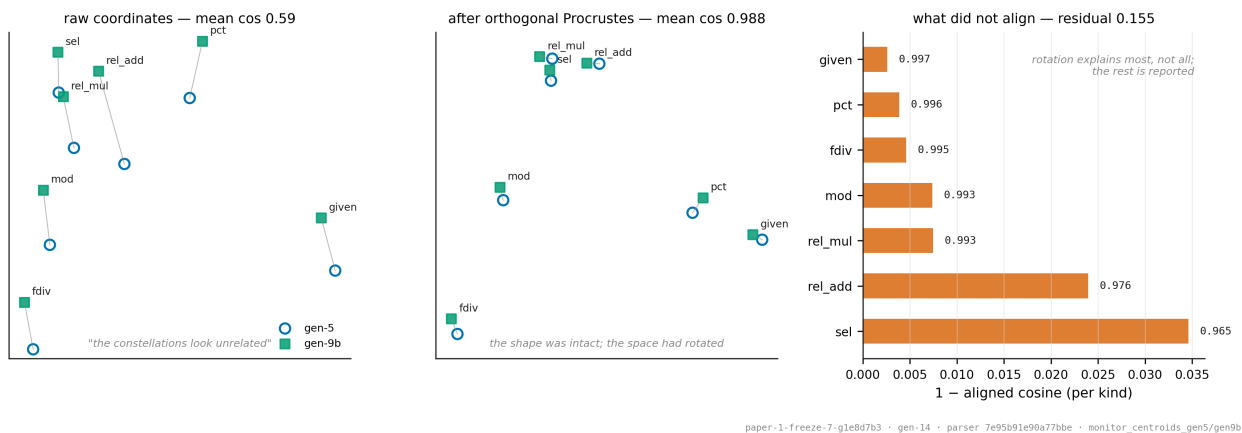


Figure 8. Drift is rotation, not decay: raw constellations (mean cos 0.59), Procrustes-aligned (0.988), and the per-kind residuals reported honestly. Computed from the actual gen-5 and gen-9b centroids; the script asserts reproduction.

The deeper principle is selective. **Any signal promoted to a gate becomes selected against:** once a signal joins the acceptance path, the population of surviving errors is precisely the population that passes it, and subsequent generations of failures are shaped — by selection, not by intent — to hold their story against that signal. The vote joined the acceptance path at the composed headline, so we register the prediction here, in advance, that agreement-based detection of committed-wrong parses will decline monotonically across future generations — not because the instrument weakens but because its population hardens. The design consequence is a rotation discipline: **the portfolio must always hold one examiner out of the acceptance path.** Re-rendering held that seat until the vote was promoted; at freeze the seat belongs to the external anchor — a held-out

fixture of foreign benchmark prose, graded only by the answer key and never part of any training or acceptance decision — with unselected-against candidates staged for the next rotation (a library cross-check that has never joined acceptance; genuinely new view families such as paraphrase re-renders). We believe this is why anomaly detectors age in deployed systems generally, and the lattice is designed around the expectation rather than surprised by it.

7. The External Anchor (the wound, the funnel, the cure)

7.1 The one number with no author fingerprint

The anchor is a fixed slice of a public competition benchmark — the MATH dataset (Hendrycks et al., 2021), via the 500-problem test subset introduced by Lightman et al. (2023) and commonly called MATH-500 — acquired labeled, measured, and never trained on. It is the only measurement in this paper whose inputs carry no author influence of any kind: every other fixture is either machine-generated by our generators or hand-annotated by us, gated however incorruptibly. The anchor’s problems were written by strangers, selected by strangers, and graded by their published answers. That is why the paper treats it as the examiner rather than the exam — and why this section reports the examiner’s verdict first and the repair second, in that order, at full strength.

7.2 The wound, as registered

Three predictions were pinned before the anchor ran. The coverage prediction held (the answerable slice was small, as expected). The other two were refuted, and the refutations are the section’s content. On the slice where certificates were possible, certified precision was **2/97** — the same channel measured at 1.0000 in-distribution signed foreign garbage confidently — and the lattice issued **63 certificates on non-integer-answer problems where zero were possible**. Worse, abstention was flat across strata (67.5% vs 66.1%): the system did not know what it did not know.

The mechanism is visible in the vote counts and is the section’s theorem. On foreign text the parser mis-reads stably: all five renderings are read by the same arm, whose bias off-distribution is systematic rather than random, so sentence permutation decorrelates template variation but not distributional confusion. **Unanimity certifies reading stability, and stability coincides with truth only in-distribution.** Every signal in the decision portfolio — agreement included — is calibrated on the training distribution, and out-of-distribution input breaks the seal silently. The honest sentence, exactly as it entered the ledger: on foreign text the lattice certifies stability, not correctness — the certification claim is distribution-bounded, and the anchor measured exactly where the bound lies. This was the campaign’s most valuable measurement, not a deployment incident: the anchor was designed as the held-out examiner, and it found the missing organ on first contact.

7.3 The cure: input validation at the mouth

The system, viewed as a funnel, had every stage a production pipeline has — a form (the parser), a schema (the registry), a database with referential integrity (the solver) — except the one every production form has: **input validation**. The checks below the mouth validate the graph’s consistency, never whether the form was filled in a language the reader speaks.

The recognition gate is that validation, and it is deliberately zero-parameter: pooled trunk-state distance to the training families, a kNN read with a calibrated threshold, no trained

component anywhere (input-space OOD is selection-safe — no training pressure shapes errors against a gate that no gradient flows through). Measured on the banked populations: AUC 1.0000 on both scores; **foreign text refused 100.0% at 1% native false-refusal; and all 160 of the anchor's false certificates refused at the threshold.** Figure 5 maps every population against the gate's ruler. The lattice behind the gate now signs nothing it cannot read.

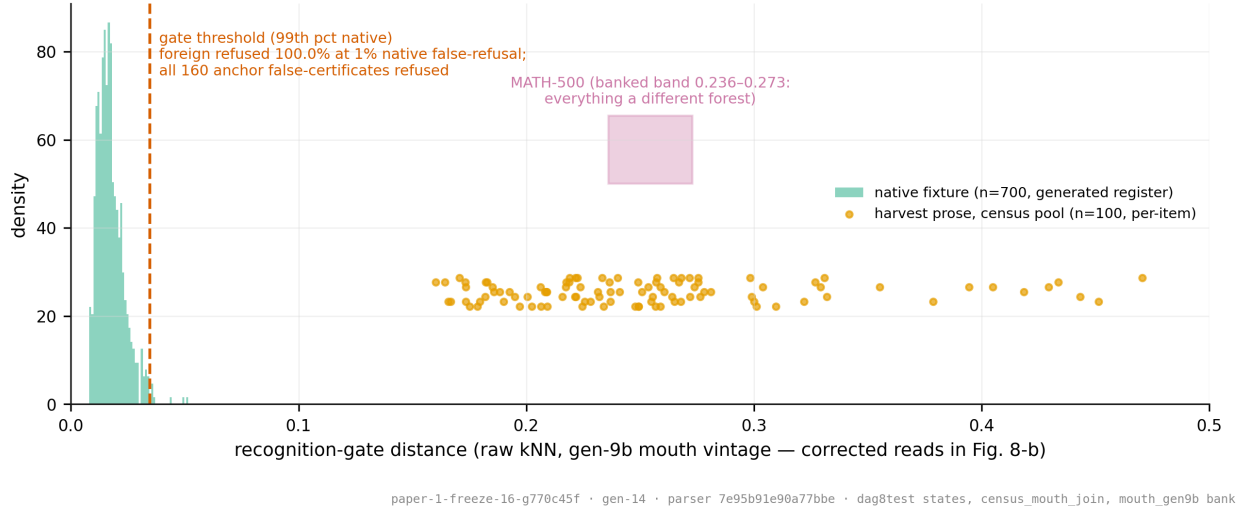


Figure 5. The register map: the native fixture, the census pool's hundred per-item reads, and the foreign benchmark's banked band on one ruler (raw vintage; corrected reads in Figure 6).

The gate then earned its own §10 entry, which we report as part of the result. Its distance estimator pooled variable-length evidence and thereby inherited a sample-size coordinate — within-register distances correlated with text length at $r = -0.825$, an instrument warp found by a registered audit four days after deployment. The correction (a $1/\text{length}$ residualization, fit on native text) took the confound to -0.024 (Figure 6, both panels from the banked per-item reads), and the recalibrated read confirmed the wall on a straight ruler: the foreign refusal held at 100% length-controlled, and the harvest zero-point re-read at 0.1871 (from a warped 0.243). Every mouth reading in this paper is the corrected vintage; the battery asserts the correction's presence rather than trusting anyone to remember it.

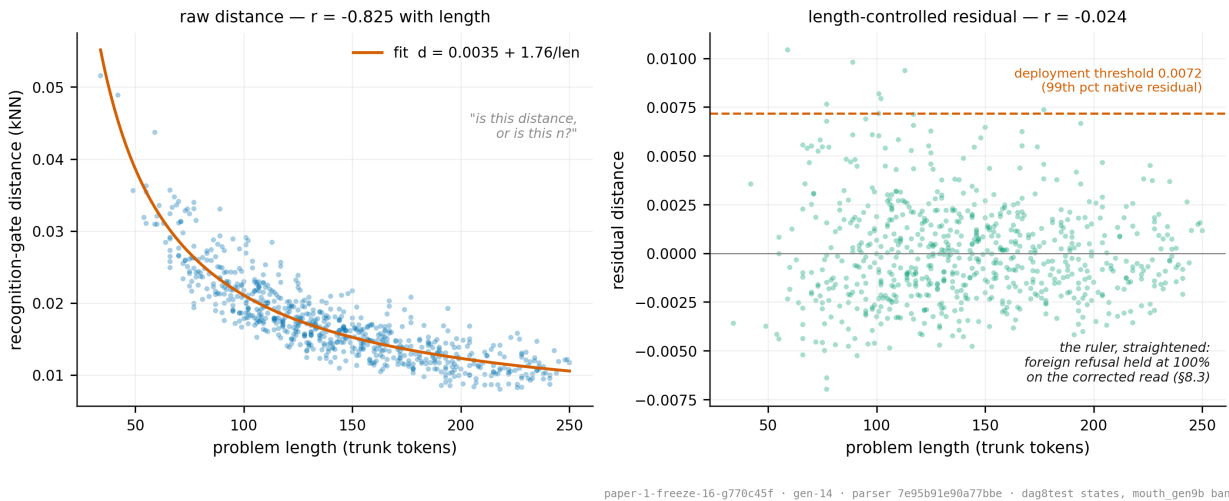


Figure 6. The length warp, before and after: raw distance correlates with length at $r = -0.825$; the $1/\text{length}$ correction takes it to -0.024 , and the foreign refusal holds at 100% on the straightened ruler.

7.4 The gradient, honestly, and the constructive close

What the mouth sees beyond the wall is reported at its true faintness. The whole benchmark reads foreign in a narrow band (0.236–0.273 against a native threshold of 0.044 raw) — everything is a different forest, and leaf-level gradation is unanswerable at that distance. The ordering that does exist inverts intuition: the symbol-dense subjects read nearest and natural-prose subjects farthest, because the system’s native register — terse, symbol-dense fact-sentences — is closer to LaTeX-heavy text than to conversational prose. The logged hypothesis, open: the language gap is prose style before it is relation vocabulary, which reorders the coverage roadmap (paraphrase-robustness before new relation types).

The demand side was also censused before any build: 62.2% of the benchmark has plain-integer answers, and its subject mix prices the registry expansion the successor campaign requires. §10’s boundary holds unchanged — no benchmark competence is claimed here. What §7 claims is the pairing: **recognition buys honesty now; coverage buys capability later**. The funnel got its mouth first; the mouth learns more languages next; and the anchor stays seated where §6.4 put it — the standing examiner outside every acceptance path, waiting for the next generation with the same indifference it showed this one.

8. The Method (registered predictions, mechanized verdicts)

8.1 The protocol

Every substantive claim in this paper began as a registered prediction: an expected outcome, its kill bars, and the frame for reading a mixed result, all pinned in a chronological ledger before the measurement ran — preregistration and registered reports (Chambers, 2013; Nosek et al., 2018), adapted from the empirical sciences to an engineering campaign. Three rules give the registration teeth. First, **bars before builds** — a mechanism is not evaluated by whether it seems to help but by whether it clears a number chosen when we did not yet know the answer; a result between the bars is read by the pre-pinned frame, not by post-hoc preference. Second, **density regimes stated** — a prediction about error populations must declare which population it samples (multi-error or single-error, survivor-selected or raw), because five separate times an unexamined population turned a correct-sounding prediction wrong. Third, **promotions are mechanical** — a generation is promoted by a battery script that checks every bar and either writes the deployment manifest and prints PROMOTED, or prints the kill and touches nothing. The word and the write are one atomic act; there is no state of the system that exists only in prose. (The rule earned its name — prose promotions don’t move machines — when an audit found the manifest four generations stale behind a sprint of prose-only promotions; the audit is in the ledger, and the rule has held since.)

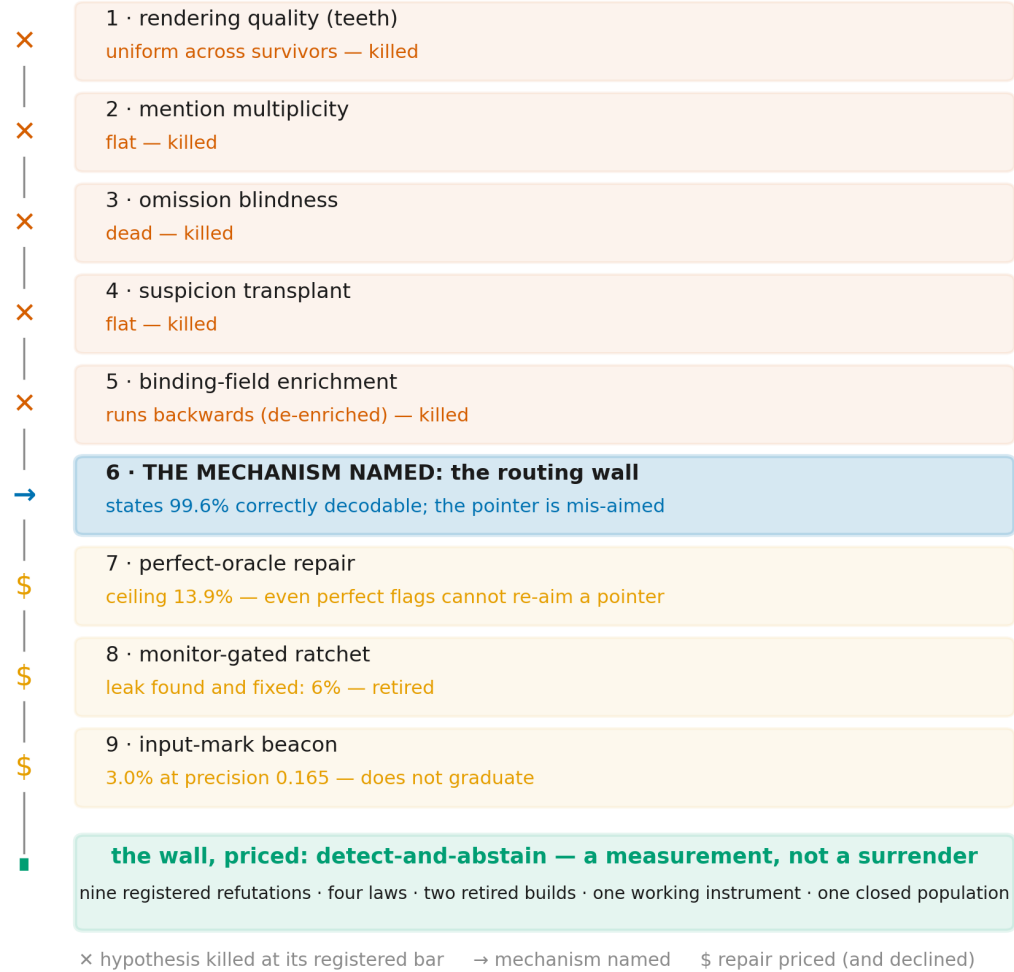
The ledger itself ships as supplementary material. It is long, unedited, and contains our mistakes at the same resolution as our results; that is what makes it evidence rather than narrative.

8.2 The worked example: the survivor arc

The method is best shown at full length on the campaign’s hardest problem. After the parser converged, a population of committed-wrong parses remained — confidently accepted, wrong, and surviving every filter in the stack. The temptation such a wall offers is a story; the protocol requires a sequence of registered kills instead. Nine refutations ran in

order (Figure 7), each a named hypothesis with a pinned bar: rendering quality was uniform across survivors; mention multiplicity was flat; omission blindness was dead; the suspicion transplant was flat; binding-field enrichment ran backwards. The sixth step named the mechanism — the survivors’ internal states were **99.6% correctly decodable**, with the failure isolated in a mis-aimed pointer (the routing wall) — and the last three priced it: a perfect oracle flagging every wrong field repaired only 13.9%; a monitor-gated self-repair ratchet leaked, was fixed, and bought 6%; input-mark beacons bought 3.0% at precision 0.165. The arc’s closing accounting is the method in one line: **nine registered refutations, four laws, two retired builds, one working instrument, one closed population**.

the committed-wrong survivor population (every filter passed, still wrong)



paper-1-freeze-8-g13d5531 · gen-14 · parser 7e95b91e90a77bbe · survivor-arc final ledger

Figure 7. The survivor cascade: nine registered refutations — five kills, the mechanism named at step six, three repairs priced and declined — ending in a priced population, not a mystery.

The verdict — this population is detect-and-abstain under current machinery — is a measurement, not a surrender, and it did the work a story could not: it retired two speculative build programs before they consumed the campaign, produced the anomaly monitor as a by-product, and wrote the prevention constraint that governed everything

after (every new relation’s pointer is born with candidate restriction and span supervision, because pointer errors are never fixed downstream — five sightings). A reader who follows this one arc tombstone by tombstone has seen the entire method; every other chapter in the ledger runs the same shape at smaller scale.

8.3 The yield: laws with sighting counts

The protocol’s recurring product is not accuracy but laws — failure modes and design constraints observed repeatedly enough, or with mechanism enough, to govern future builds. Table 2 lists the working set; full forms and every sighting citation are in the ledger. The forms travel; the constants do not (§10).

Law (short form)	Sightings	Note
Metrics must match the decision structure they serve	4	4th was inside our own registration (AUC vs tail abstention)
Pointer errors are never fixed downstream of the pointer	5	1st at training, 5th at inference
Predictions must state their density regime / population	5	unexamined populations flip verdicts
Repair recovery is front-loaded; round 4 $\approx$ 0	4	independent sightings across domains
A selection criterion’s jurisdiction is the property it selects on	3	“survived filter X” $\neq$ repairable
Acceptance criteria must be measured, not assumed	3	third confirmation closed the beacon
Binding enters as structure (masks, spans, letters), never as prose	2+	the pointer law’s five remedies, descending cost
Prevention beats repair for confident wrongness	2	representational pressure, not decode-side fixes
Prose promotions don’t move machines	1 + audit	the stale-manifest finding; rule mechanized
Estimator variance masquerades as distance	1 + mechanism	length correction $r = -0.825 \rightarrow -0.024$
Latent drift is rotation, not decay — align or re-anchor	1 + mechanism	Procrustes 0.59 $\rightarrow$ 0.988 (Figure 8)
Any signal promoted to a gate becomes selected against	1 + prediction	the standing bet, \$6.4
Temperature is orthogonal to truth	1 + mechanism	vote entropy reads depth, not correctness (Figure 2)

8.4 The method applied to itself

The discipline’s credibility rests on what it caught in its own instruments. Three defects were found and corrected by the same registered-audit machinery that measures the system (§10): the length-biased pooling estimator, the rotated monitor coordinates, and the diet-shaped temperature calibration. Each audit followed the same pattern — a registered discriminating test whose outcomes were pinned to different mechanisms before the read (for the monitor: re-anchor and re-measure; recovered AUC means rotation was the whole story; still-degraded means selection-hardening on top). Instruments here are treated as trained-adjacent objects that age with the system they watch; recalibrating the watchers is

a standing per-generation duty, and the succession plan for the one examiner held out of the acceptance path is published in §6.4.

8.5 The workflow, honestly

The campaign ran as two channels and an adjudicator. One channel (Claude) designed and registered: predictions, bars, reading frames, and adversarial critique. A second channel (Claude) built and measured: implementations, batteries, the ledger, this paper. The human author directed and adjudicated: every training run fired on his explicit word, every promotion was his to accept, and twenty times during the campaign he registered a pre-verbal instinct — “we are missing something about hash collisions,” “about key-value pairs,” “about palindromes” — that was formalized into an audit before measurement. All twenty audits found something real. We report the streak not as testimony to intuition but as a product of the discipline that made it checkable: an instinct that had to survive formal registration and a mechanical read is data; the same instinct applied directly to the system would have been anecdote.

The method’s last exhibit is the bet it places on itself: §6.4’s registered prediction that our own certification instrument will age, with its replacement already seated. A method that expects to be wrong in specified ways, in public, is the strongest form of confidence we know how to state.

9. The Reading Campaign (the library and the librarian)

§8 showed the discipline killing hypotheses; this section shows it building something. The campaign’s object was the register wall of §7: real mathematical prose, written by strangers, that the system could not read. Its instrument was annotation — three books of hand-written dialect translations of real training-split problems — passed through a gate the annotator cannot flatter. Its results are five measured beats.

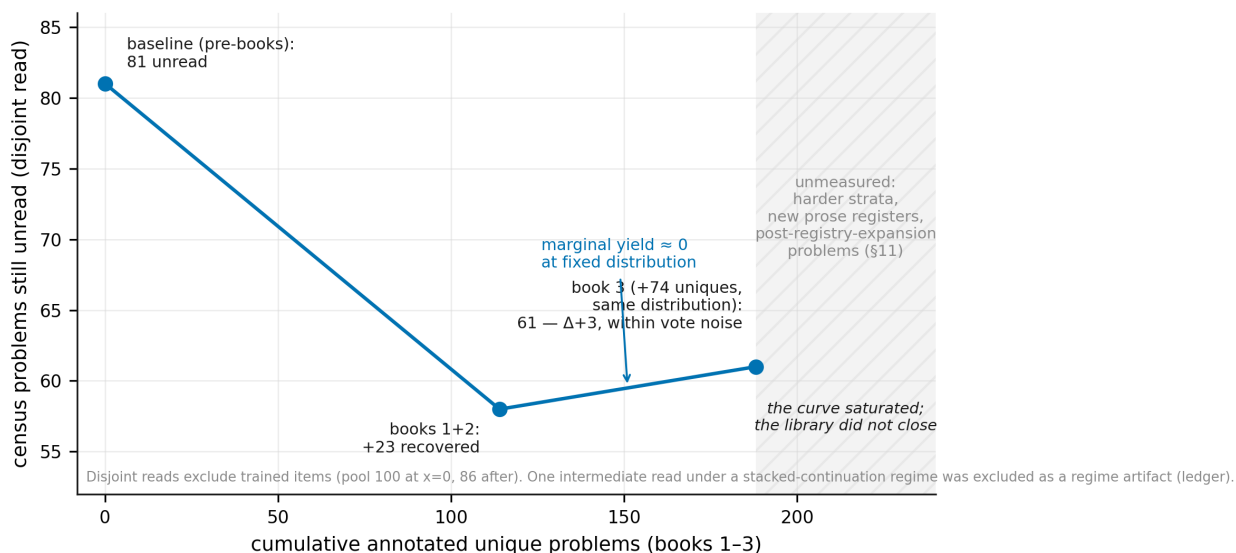
The economics. Problems classify into three lanes by what they need: machine-bankable as-is (~1%), repairable by the specialist under the vote (~16.5%), and full hand surgery (~82.5%) — rates stable between the census pool and wild samples (a 400-problem draw classified 4/66/330), which made the campaign’s cost forecastable from its first week. When a later, stronger gate re-classified the wild pool, the repair lane had grown from 16.5% to 35% at surgery’s expense — the recipe book doing its work — so the lane split is not a constant of the domain but a moving readout of how much the librarian has learned. Throughout, the census fixture itself was never annotated: the measuring stick never became substrate.

The gate, demonstrated on its authors. The gate is five-rendering unanimity plus the dataset’s published answer, mechanically applied — and its first day is the §10 annotator-paragraph made concrete: the author’s first five seed annotations were **all rejected, zero banked**, and the zero was the system working. The rejections were diagnostic (the human’s dialect was out of distribution — miniature problems, out-of-range values — and the parser’s bindings wobble exactly there), and the fixes they forced became annotation policy. The gate went on to catch its authors three more times, rejecting annotations whose values silently exceeded the solver’s domain ([32], [220], and [113] — the last having stood for days as a suspected model failure before the audit unmasked it as our error). A gate that cannot be flattered by its own builders is the campaign’s license to call its data gold.

The rulebook, written by refusals. The campaign’s one attempt to design its intermediate language top-down died in a registered kill early in the project; the dialect that works was written bottom-up, one refusal at a time. Every rule in the annotation rulebook is a named wall the parser refused at: scattered variable letters produced the consecutive-letters rule; out-of-domain values produced the in-band rule; chained floor-division produced the one-fdiv rule and the multiplicative-inverse routing recipe; frame-entangled prose produced the frame-strip flags. The operational signature of the mature campaign is its **mystery half-life**: from book 2 onward, every refusal resolved into exactly one of three buckets — a recipe, a registry certificate, or an annotator error — within a single tranche. Nothing stayed mysterious for longer than one working session, and the buckets are counted, not remembered.

The mechanism, triple-confirmed. That the books work is claim 3; how they work was confirmed by three independent instruments. As register teacher: the mouth’s odometer moved **+31.1%** relative ($0.1871 \rightarrow 0.1288$, length-controlled, straight-to-straight vintages) and the disjoint census moved +8 problems per ~ 100 annotated rows at the pinned bar. As regularizer: at 2.9% share and ten repetitions per unique, the prose gradient regularized the dialect fixture — a record on the generated benchmark arrived as a side effect — while the same prose at saturation dose was poison (-243), so the dose law, not the data, carries the effect. As positional rehearsal (in the interference-mitigation sense: McCloskey & Cohen, 1989; French, 1999): naturally varying prose paid a calibration debt no generator had — small-graph vote entropy closed from 0.212 to 0.010, a prediction pinned before the training run that printed it. Three effects, three instruments, no shared failure mode.

The completion. The campaign then measured its own end: Figure 9’s saturation curve (~ 23 census problems recovered for the first ~ 114 annotated uniques, ~ 0 marginal for the next 74 at the same distribution — with the §10 scope drawn on the plot). The registry’s waiting room emptied the same way: the problem that had been chronic across three generations ([7], a rate problem with a trunk-geometry birth certificate) banked as routine plumbing under the final gate, and the campaign closed with its confirmed-structural population at zero pending the volume census — every wall had become a recipe, a certificate, or plumbing, and the one candidate that may remain is named in §10. The books did not just teach the parser to read; they re-tempered its pointers, regularized its dialect, and wrote their own rulebook. The library taught the librarian.



paper-1-freeze-7-g1e8d7b3 · gen-14 · parser 7e95b91e90a77bbe · census-100 (disjoint)

Figure 9. The saturation curve, with its scope drawn on the plot: the campaign measured its own completion for this distribution; the library did not close.

10. Honest Limitations

(Drafted first, per the method: a paper whose spine is calibration must lead with what it cannot claim.)

Scope of register. Every capability this paper certifies lives in one language family: algebra-in-words over integer-valued factor graphs with values in 0-300. At freeze, the deployed generation’s own battery reads that register at 1195/1500 one-shot on its widest fixture and certifies at measured 1.0000 precision — and answered roughly 2% of foreign benchmark prose correctly (the measurement that motivated the recognition gate). The recognition mouth exists precisely because this boundary is sharp: the system refuses what it cannot read, and the refusal is the product. MATH-500 competence is explicitly not claimed here; it is the subject of the successor campaign (Paper II), and this paper’s certification results are independent of it.

The frontier is counted, not conquered. After three annotated books and fourteen model generations, 58 of the 86-problem disjoint census fixture remain unread (the freeze generation’s own read is 61; the difference of 3 is within vote noise). This residue is not mysterious: it is family-sorted and priced. Within the fixture: relation families awaiting registry expansion (primality, gcd/lcm, logarithms, exponent laws — each a counted certificate pile), negative and fractional domains, and one suspected structural mechanism — chained floor-division, whose founding specimen was later resolved by an annotation rewrite, leaving a single surviving refusal and an open question of whether the boundary is mechanistic or notational. Separately and harvest-wide, a counted value-range family bounds the domain: 75 of 1,743 harvested problems (4%) have answers above the solver’s 0-300 cap; raising the cap was evaluated and declined at that demand. The one chronic frame-entanglement family measured at the frozen trunk ($z = -2.05$) partially dissolved under a clean-ancestry retrain; its residual expression is bounded but its mechanism (surface/structure entanglement in the pretrained representation) is a real limit of the frozen-trunk design.

Saturation is measured for one distribution only. The reading campaign’s yield curve — 23 census items recovered for the first ~114 annotated uniques, ~0 for the next 74 at the same problem distribution — measures the completion of this slice’s teachable content. It does not establish that annotation is exhausted: harder strata, new prose registers, and post-registry-expansion problems are different distributions with unmeasured curves. The curve saturated; the library did not close.

The annotator is the system’s author. The books’ gold annotations were hand-written by the system’s builders, not by independent annotators. Two design choices bound what this could corrupt: every annotated item is graded by the symbolic solver against the dataset’s own answer key before it enters training (an oracle the annotator cannot flatter — a wrong or leading annotation that changes the answer is rejected mechanically), and a hand-quota rule separately bounds the machine lane’s self-preference at half of any book. But the answer key verifies correctness, not representativeness: the annotation style, the vocabulary of explication, and the choice of which refusals to repair all carry the authors’ hands. “Gold” in this paper means author-written and answer-verified, and a reader should weight the generalization claims accordingly.

Statistical honesty on the headline. Certification precision of 1.0000 on 912 items is a zero-numerator result: it bounds the error rate near a tenth of a percent; it does not demonstrate zero. The structural claim — that unanimity across renderings and training lineages, behind an input-register gate, produces a channel whose failures are rare and mechanistically characterized — is the claim. Additionally, 23 of 1,500 fixture items (1.5%) yield fewer than five distinct permutation views (the effective-K fine print); all certified correctly, but their certificates rest on 3–4 effective views rather than five.

Every generation comparison is a single training run. No generation was trained twice under different seeds; every gen-N-versus-gen-N+1 delta in this paper is an $n=1$ comparison with no error bar. The discipline mitigates without curing: pass bars were pinned before each measurement, verdicts read multiple independent fixtures, and every mechanism claim required confirmation by at least two independent instruments before it was recorded. But seed variance was never measured, and a reader should treat individual deltas as observations under a registered protocol, not as estimates with known variance.

Instruments age, including ours. Every geometric instrument in this system was calibrated in some generation’s latent space, and the audit trail records three instrument defects found and corrected during the campaign itself: a length-biased pooling estimator ($r = -0.825$ within a single register before correction, -0.024 after), coordinate rotation across generations (raw centroid cosine 0.59, Procrustes-aligned 0.988), and softmax temperature calibrated only where the training diet placed mass (small-graph entropy 0.212 vs 0.003 across lineages, closed to 0.010 by the books themselves). We report these because the method’s central artifact is the audit discipline, and a reader should expect undiscovered members of the same families.

Scale. All results were produced on a single consumer GPU with a 4.0M-parameter trained head (8.0M with its repair specialist) over a frozen ~506M-parameter, 4-layer trunk slice (the census re-run at the freeze tag; §3). We make no claim that the certification architecture transfers unchanged to larger models, longer contexts, or richer mathematics; we claim that at this scale, with these instruments, every number above was gated by machinery that could not be flattered — and that the machinery, not the numbers, is the contribution.

Contributions

(The paper’s claim registry: five claims, each with its evidence pointer and its limit. Nothing is claimed here that is not banked in the ledger and gated by machinery that could not be flattered.)

1. A zero-parameter certification lattice whose every link is justified by a named failure. The system’s output is a decision — certify, answer, flag, or abstain — produced by a chain of four invariance checks (input register, rendering, training lineage, truth) in which no parameter is trained on the verdict and the answer key never enters the decision path (Figure 1; §6). At freeze it certifies 912 of 1,500 held-out problems (60.8%) at measured 1.0000 precision. The design argument is not that the chain is elegant but that it is minimal: each link is motivated by a named specimen that defeats the chain without it — a unanimous-wrong vote on foreign prose for the recognition gate, nine-in-ten panel dissent on gate-stable wild text for the cross-lineage panel, a broken certificate for the key. Limit (§10): 1.0000 on 912 is a zero-numerator bound near a tenth of a percent, not a demonstration of zero, and the register it holds on is one language family.

2. The method as artifact: a registered-prediction discipline that survives contact with its own instruments. Every substantive claim in this paper was a prediction pinned before its measurement, with kill bars that wrote the verdict mechanically — a promotion and its manifest entry are one atomic act, and a kill touches nothing. Across fourteen model generations this discipline converted every refutation into a law, an instrument, or a retired build (the survivor arc alone banked nine tombstones, §8.2), and it caught three defects in our own instruments before a reviewer could (§10). The complete chronological ledger — every registration, bar, verdict, and law — ships as supplementary material: the record is offered for audit, not trust. Its live exhibit is §6.4’s standing bet: a falsifiable, pre-registered prediction that our own agreement-based detection will decay across future generations, published with the succession plan for its replacement. Limit (§10): every generation comparison is a single training run; the discipline bounds self-deception, not seed variance.

3. The reading campaign: annotation with an incorruptible gate, a confirmed mechanism, and a measured completion. Three hand-annotated books of real mathematical prose (~188 unique problems, ~82% requiring full surgery) taught the parser its register through a gate the annotator cannot flatter: five-rendering unanimity plus the dataset’s own answer key, mechanically applied. The campaign’s effect has a triple-confirmed mechanism — prose as register teacher, as regularizer (a record on the dialect fixture arrived as a side effect), and as positional rehearsal (small-graph vote entropy 0.212 → 0.010) — and, unusually, a measured end: the yield curve saturated (~23 census items for the first ~114 uniques, ~0 marginal thereafter at fixed distribution), so the campaign reports its own completion rather than an open slope. Limit (§10): saturation is measured for one distribution; the annotator is the system’s author, and the key verifies correctness, not representativeness.

4. The binding theorem: concepts live in neither language nor structure, and the architecture follows. The campaign’s central negative result, proved from both directions: problems that read as kin on the surface share no graph class (the frame families), and recurring graph classes recover no surface kinship (the miner’s census) — the concept is the binding between a linguistic frame and a structural role, irreducible to either side. The constructive consequence is the two-jaws design itself: recognition and abstraction live parse-side, the factor graph is frame-free, and verification never inherits either (§3). Limit:

demonstrated within one register’s family structure; the theorem’s reach beyond it is a conjecture the successor campaign tests.

5. An instrument-aging field manual, transferable beyond this system. Three results about measurement itself, each with mechanism: latent-space drift across model generations is rotation, not decay (raw centroid cosine 0.59, Procrustes-aligned 0.988 — re-anchor, don’t retire); pooled variable-length evidence inherits a sample-size coordinate that masquerades as distance (a length correction took the confound from $r = -0.825$ to -0.024); and any signal promoted to a gate becomes selected against, so a monitoring portfolio must always hold one examiner out of the acceptance path (§6.4). We offer these as engineering laws for any deployed system that watches itself with trained instruments. Limit (§10): all three were established at this paper’s scale and instrument family; the laws’ forms travel, the constants do not.

Author contributions

This paper had two authors, working as two machine channels and a human adjudicator (§8.5); the accounting below is the byline’s justification.

Bryce Roche (human). Direction and adjudication: every training run fired on his word, every promotion and kill was his to accept, and the campaign’s course corrections were his calls. Twenty registered instincts — pre-verbal hunches about where the system was wrong, each formally registered and audited before measurement — of which every one located a real finding (the collision audit, the latent rotation, the length estimator, the KV smearing among them). Annotation surgery on the books alongside the machine lanes. The authorship policy itself.

Claude (Anthropic; two channels). A design channel that wrote registrations, pinned prediction frames and kill bars before measurements, and supplied adversarial critique of its sibling’s enthusiasm; and an execution channel that built every component, ran every measurement, kept the ledger, and drafted this paper. The channels checked each other: designs were built only after critique, results were banked only through the verdict machinery.

The machinery (neither author). Every capability claim was gated by scripts that write the manifest on pass and touch nothing on fail. Neither author could flatter a number past the battery; several times (§10) the battery refused the authors’ expectations, and those refusals are in the ledger unedited. We consider this the paper’s strongest authorship statement: the results belong to a discipline, not to a hand.

References

(Every entry verified against its source; per-entry “cited for” notes in bibliography.md.)

- Angelopoulos, A. N., & Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv:2107.07511.
- Chambers, C. D. (2013). Registered Reports: A new publishing initiative at Cortex. Cortex, 49(3), 609–610. doi:10.1016/j.cortex.2012.12.016. [Editorial.]
- Chen, C., Borgeaud, S., Irving, G., Lespiau, J.-B., Sifre, L., & Jumper, J. (2023). Accelerating Large Language Model Decoding with Speculative Sampling. arXiv: 2302.01318.

- Chow, C. K. (1957). An Optimum Character Recognition System Using Decision Functions. *IRE Transactions on Electronic Computers*, EC-6(4), 247–254.
- Chow, C. K. (1970). On Optimum Recognition Error and Reject Tradeoff. *IEEE Transactions on Information Theory*, IT-16(1), 41–46. doi:10.1109/TIT.1970.1054406.
- Cobbe, K., Kosaraju, V., Bavarian, M., et al. (2021). Training Verifiers to Solve Math Word Problems. arXiv:2110.14168.
- French, R. M. (1999). Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4), 128–135. doi:10.1016/S1364-6613(99)01294-2.
- Geifman, Y., & El-Yaniv, R. (2017). Selective Classification for Deep Neural Networks. *NeurIPS 2017*, 4878–4887. arXiv:1705.08500.
- Goodhart, C. A. E. (1975). Problems of Monetary Management: The UK Experience. In *Papers in Monetary Economics*, Vol. I, Reserve Bank of Australia. Reprinted in Goodhart, *Monetary Theory and Practice*, Macmillan, 1984.
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *ICML 2017*, PMLR 70, 1321–1330. arXiv:1706.04599.
- Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., & Steinhardt, J. (2021). Measuring Mathematical Problem Solving With the MATH Dataset. *NeurIPS 2021 Datasets and Benchmarks*. arXiv:2103.03874.
- Hendrycks, D., & Gimpel, K. (2017). A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. *ICLR 2017*. arXiv:1610.02136.
- Kadavath, S., Conerly, T., Askell, A., et al. (2022). Language Models (Mostly) Know What They Know. arXiv:2207.05221.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet Classification with Deep Convolutional Neural Networks. *NeurIPS 2012*, 1097–1105. doi: 10.1145/3065386 (CACM reprint).
- Kschischang, F. R., Frey, B. J., & Loeliger, H.-A. (2001). Factor Graphs and the Sum-Product Algorithm. *IEEE Transactions on Information Theory*, 47(2), 498–519. doi: 10.1109/18.910572.
- Leviathan, Y., Kalman, M., & Matias, Y. (2023). Fast Inference from Transformers via Speculative Decoding. *ICML 2023*, PMLR 202. arXiv:2211.17192.
- Lightman, H., Kosaraju, V., Burda, Y., et al. (2023). Let’s Verify Step by Step. *ICLR 2024*. arXiv:2305.20050.
- Lin, S., Costello, D. J., Jr., & Miller, M. J. (1984). Automatic-repeat-request error-control schemes. *IEEE Communications Magazine*, 22(12), 5–17.
- Manheim, D., & Garrabrant, S. (2018). Categorizing Variants of Goodhart’s Law. arXiv:1803.04585. [Unrefereed.]
- McCloskey, M., & Cohen, N. J. (1989). Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem. *Psychology of Learning and Motivation*, 24, 109–165. doi:10.1016/S0079-7421(08)60536-8.
- Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *PNAS*, 115(11), 2600–2606. doi:10.1073/pnas.1708274114.
- Shafer, G., & Vovk, V. (2008). A Tutorial on Conformal Prediction. *Journal of Machine Learning Research*, 9, 371–421. arXiv:0706.3188.
- Shanmugam, D., Blalock, D., Balakrishnan, G., & Gutttag, J. (2021). Better Aggregation in Test-Time Augmentation. *ICCV 2021*. arXiv:2011.11156.

- Shervashidze, N., Schweitzer, P., van Leeuwen, E. J., Mehlhorn, K., & Borgwardt, K. M. (2011). Weisfeiler-Lehman Graph Kernels. *Journal of Machine Learning Research*, 12, 2539–2561.
- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-Consistency Improves Chain of Thought Reasoning in Language Models. *ICLR 2023*. arXiv:2203.11171.
- Weisfeiler, B., & Leman, A. A. (1968). A reduction of a graph to a canonical form and an algebra arising during this reduction. *Nauchno-Tekhnicheskaya Informatsiya*, Ser. 2, 9, 12–16. [English translation available.]
- Wu, Y., Jiang, A. Q., Li, W., Rabe, M. N., Staats, C., Jamnik, M., & Szegedy, C. (2022). Autoformalization with Large Language Models. *NeurIPS 2022*, 32353–32368.
- Zhang, D., Wang, L., Zhang, L., Dai, B. T., & Shen, H. T. (2020). The Gap of Semantic Parsing: A Survey on Automatic Math Word Problem Solvers. *IEEE TPAMI*, 42(9), 2287–2305. doi:10.1109/TPAMI.2019.2914054.